Reserve-Requirement Shocks and Loan Growth: Data and Main Specification

Reference note for Table 3

1 Overview

Research question: Does deposit funding lead to more credit line supply relative to term loans?

Main Regression:

$$\Delta \log(Y_{f,b,c,t}) = \alpha_{f,c,t} + \beta_c X_{b,t} + \varepsilon_{f,b,c,t}$$

where

- Index: f firm, b bank, $c \in \{T, C\}$ credit type, t time (month)
- Y: credit volume
- X: reserve requirement shock (bank level)

Table 3 is the main result. It reports four regressions. Each one asks the same question. When the central bank raises reserve requirements, do the banks that are more exposed to the change cut lending more than less-exposed banks that lend to the same firm in the same month?

The regressions differ in two ways. First, they use two loan types: credit lines (columns 1 and 3) and all other credit (columns 2 and 4). Second, they use two measures of the reserve-requirement change: the minimum legal rate (columns 1 and 2) and the marginal rate (columns 3 and 4).

The outcome is the monthly log growth of a firm's balance at a bank. The variable of interest is a bank-level shock. It is the change in the reserve rate, weighted by the bank's fixed 2009 deposit mix across domestic and foreign currency. Every regression absorbs firm-by-date fixed effects. So the coefficient compares two banks lending to the same firm in the same month.

2 Institutional background

2.1 Reserve Requirement Details

Peru's reserve requirement framework is built around two main rates that the BCRP sets separately for soles and U.S. dollar obligations. The first is the minimum legal rate, a floor on the reserve a

bank must hold as a share of its total reservable obligations. Under a single-rate regime, this rate applies to every reservable obligation. Under a base-plus-marginal regime, it operates as a floor.

The second is the marginal rate, which applies only to obligations above a bank-specific base. The base is the bank's average daily reservable obligations in the general regime over a single reference month named in each circular (for instance, circular XXX from September 2021 set the reference month as July 2021 for the domestic currency reserves, and the marginal rate was in force from September 2021 to 2024). This design lets the central bank discourage balance sheet growth in a given currency without raising the reserve on existing obligations.

The rules around the base have changed repeatedly across circulars. The reference month has been reset from time to time. Banks without a history, such as new entities, have no base: the circular instead sets a flat rate up to a specified amount and the marginal rate on the remainder, while in mergers the banks' bases are added together. As a result, the average reserve rate a bank faces lies between the floor and the marginal rate, moving toward the latter as obligations grow beyond the base. Because the base rate depends on each bank's own reserve history in the reference month, the average rate cannot be reconstructed from the circulars alone.

3 Data

3.1 Sources

The analysis combines four datasets. Table 1 lists them.

Table 1: Datasets used in the main regression

Dataset	File	Content
Firm credit	`muestra_anonima_base2015.parquet`	Pseudonymised extract of the SBS credit register (RCC). One row per firm, lender, currency, segment, product, month. Sep 2010 to Dec 2023. 267,030 rows, 1,000 firms, 83 lenders.
Bank deposits	`Reporte(01-2001 - 09-2022).csv`	SBS statistical series. Deposits by institution and currency, in thousands of soles, monthly. Used only for the December 2009 deposit shares.
Reserve rates	`bcrp_encaje/encaje_rates_monthly.csv`	Reserve-requirement rates rebuilt by hand from 186 BCRP circulars. Monthly panel, one column per rate.
Lender names	`RCC_empresas_supervisadas_2026.xls`, `bank_map_manual.csv`	RCC entity code to institution name, and a hand-curated map from names to a common bank key (`bank_standardized`).

3.2 Credit data

The credit data come from the SBS credit register (Reporte Crediticio Consolidado, RCC). The register records, each month, the balance that each supervised lender holds on each debtor. The extract is pseudonymised: firm identifiers are anonymous codes.

Variables used. The regression uses six columns: the report date, the firm identifier, the lender code,

the currency code, the product, and `credito_vigente`. `credito_vigente` is the outstanding balance of performing credit (SBS account 14x1). It is a drawn balance. The file also holds restructured, refinanced, past-due and judicial-collection balances, and undrawn commitments. None of these enters the main table.

Sampling. The current sample was constructed by a random draw of 1,000 firm ID's from all active firms in December 2015. Once the firms were selected, all of their loans were tracked over the period from Sep 2010 to Dec 2023.

3.3 Deposit shares

SBS series on deposits by institution and currency. The current implementation covers only banks and "finance companies", monthly, from January 2001 to September 2022.

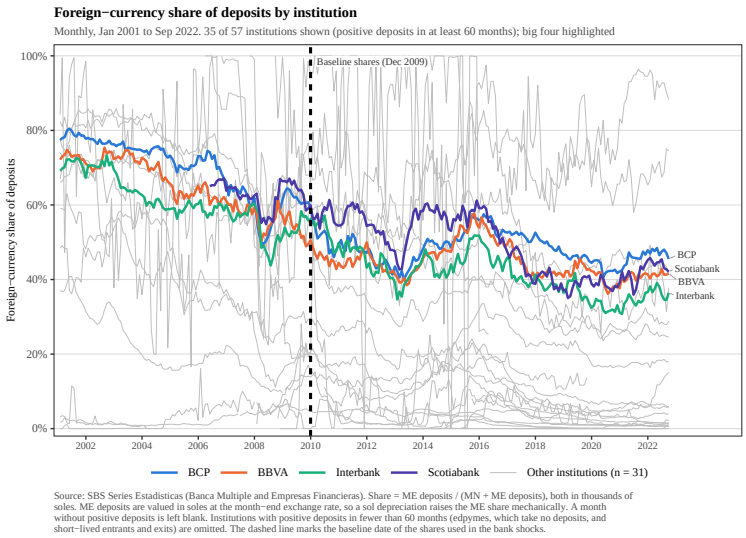


Figure 1

For each institution and month, we construct the foregin-currency deposit share.

The regression uses **December 2009** shares, held fixed. Section 6 explains why. At December 2009, 19 institutions have valid shares. Table 5 in the appendix gives the shares for the banks that enter the regressions.

3.4 Reserve-requirement rates

The BCRP does not publish a rate series.[1] The rates were rebuilt from the text of its circulars. Each circular was read and each rate change coded by Claude: effective date, currency, rate type, new value. The result is a set of monthly series (see Figure 2)

[1] Working on getting it internally.

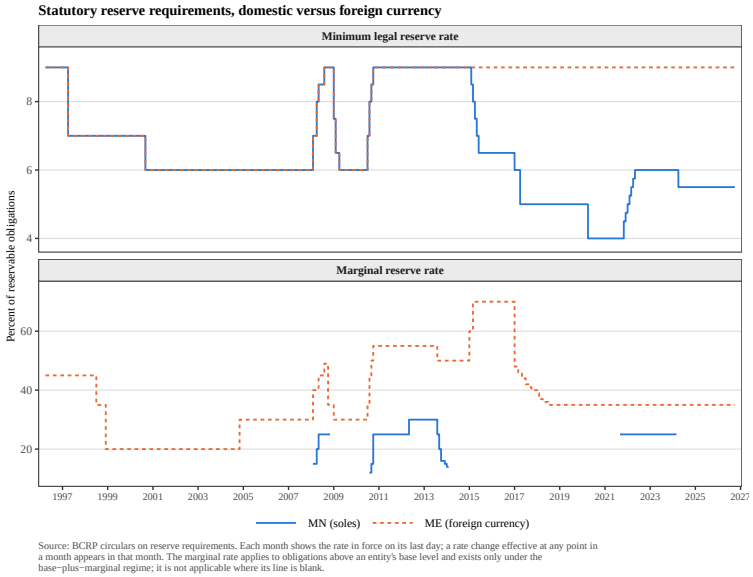


Figure 2

4 Bank Shocks

Shift-share design.

The shock for bank b in month t and rate parameter j is

$$X_{bt}^{j} \;=\; s_{b,0}^{dol}\,\Delta RR_{t}^{dol,j} \;+\; s_{b,0}^{sol}\,\Delta RR_{t}^{sol,j}, \tag{1}$$

where $j \in \{\text{min}, \text{mar}\}$. Here $\Delta r_t^{c,j} = r_t^{c,j} - r_{t-1}^{c,j}$ is the month-end change in the rate on currency c, in percentage points. The shares $s_{b,0}^{c}$ are the bank's deposit shares at the baseline date, December 2009. The weights do not change over time.

A shock of +1 means that the bank's deposit-weighted reserve rate rises by one percentage point. A positive shock is a tightening.

Equation (1) is an exposure proxy. It assumes that a rate change applies to the bank's deposits in proportion to the deposit currency mix. For the minimum legal rate, this is close to the actual mechanism. For the marginal rate it is rougher.[2] The marginal rate applies only to increases above a base level, and the shock ignores each bank's base and deposit growth.

A bank-level shock built from predetermined exposure. Every bank faces the same BCRP rates. The rates are set in response to aggregate conditions. Their time variation is common to all banks, and the date effect absorbs it. What differs across banks is exposure, and exposure comes from the deposit currency mix. A bank with mostly foreign-currency deposits is hit hard by a change in reserve requirement for foreign currency deposits. A bank with mostly soles deposits is the reverse.

December 2009 shares. The weights are fixed at a date that is before the first credit month (September 2010). It is also before the 2010 tightening cycle: the MN minimum legal rate rose from 6% to

[2]We can do better by using the bank-level required amount to complement how we construct the exposure measure (potentially backing out the baseline used for the marginal rate. Also, we are looking to get more detailed data internally at the BCRP.

9% between July and October 2010. So banks could not have chosen their deposit mix in response to the changes they are weighted by.

5 Credit Classification

For the regression shown in this document, the classification of credit as term loan or credit line is purely based on product type.

One feature of the regression data shapes the credit-line sample. Letters of credit, guarantees, multi-lines and other lines have a zero performing balance in every record. They are contingent or undrawn exposures, not drawn credit. So the credit-line regressions rest on **credit cards and current-account advances only**. Together these two products are about 1% of drawn balances in the 2015 cross-section. In the aggregated data, zero balances are dropped when growth is computed (Section 6.2).

5.1 Credit lines versus other credit

The credit register has no flag for credit lines. Two pieces of information are available. Drawn balances sit in accounts 14xx (`credito_vigente`). Undrawn limits sit in accounts 72xx (`linea_no_utilizada`). Both are tied to a product. The classification is therefore based on the product.

Six products are treated as credit lines: credit cards, current-account advances, letters of credit, guarantees, multi-line facilities and other lines. Everything else is "other credit". This includes loans, leasing, factoring, discounting, trade finance and overdrafts.

Table 2 shows the share of total credit in the sample for each product type.

Credit-line products hold little drawn credit and a lot of undrawn credit. Under the product-based classification, the credit-line products hold 1.1% of drawn balances at the end of 2015. They hold 37.1% of unused lines. The main reason for this is that some product types, such as multi-line facilities, are aggregates of the unused credit from a set of products. This is a concept for bank reporting to SBS.

Only eight products ever carry undrawn lines. Over the whole extract, the products with positive unused amounts are the six credit-line products plus loans and discounting. Discounting holds under 0.5% of unused lines at the end of 2015, so it is left in other credit.

A reporting break in June 2018. The extract has a break at 30 June 2018. Unused amounts jump between May and June 2018 for multi-line facilities (about 17-fold), guarantees (about 5-fold), discounting (about 21-fold) and letters of credit (about 1.75-fold). Almost all the growth comes from one bank BCP, which is the biggest bank in Peru. Other lines appear for the first time. Loans also rise, by about 28%. The break matters for any measure built on undrawn lines.

Loans ("prestamos") are the problem case. Loans hold 51.2% of drawn balances and 62.4% of unused lines. The revolving field does not separate them. Of the 62.4% of unused lines, almost all sit in loans whose revolving status is not identifiable. Loans flagged as revolving are only 0.8% of drawn balances and 0.04% of unused lines. Loans flagged as non-revolving are 19.7% of drawn balances and carry no unused lines. That group then contains some line-type commitments (see Section 9).

Table 2: Product classification: drawn balances and undrawn lines, December 2015

Product	Class	Revolving status	Drawn balance (%)	Unused lines (%)
Credit cards	Line	not identifiable	0.26	1.76
Current-account advances	Line	revolving	0.00	0.00
		not identifiable	0.86	11.44
Guarantees	Line	revolving	0.00	0.13
		not identifiable	0.00	13.39
Letters of credit	Line	not identifiable	0.00	3.37
Multi-line facilities	Line	not identifiable	0.00	6.99
Subtotal, credit-line products			1.12	37.08
Loans	Other	revolving	0.80	0.04
		non-revolving	19.73	0.00
		not identifiable	30.68	62.40
Discounting	Other	revolving	0.00	0.01
		not identifiable	1.40	0.47
Factoring	Other	not identifiable	3.08	0.00
Financial leasing	Other	not identifiable	14.97	0.00
Foreign trade finance	Other	not identifiable	10.91	0.00
Lease-back	Other	not identifiable	2.67	0.00
Other credits	Other	not identifiable	10.21	0.00
Project finance	Other	not identifiable	2.33	0.00
Credit to consolidable entities	Other	not identifiable	1.79	0.00
Real-estate credit	Other	not identifiable	0.19	0.00
Current-account overdrafts	Other	not identifiable	0.09	0.00
Income-generating property	Other	not identifiable	0.02	0.00
Credits to be settled	Other	not identifiable	0.00	0.00
Credit granted, not disbursed	Other	not identifiable	0.00	0.00
Other mortgage credit	Other	not identifiable	0.00	0.00
Subtotal, other credit			98.87	62.92
Total			99.99	100.00

Notes: Source: `output/tables/product_table.csv`, produced by `code/credit_classification.R` on the v2 extract. Entries are shares of the extract's total drawn balance (`credito_vigente`) and total unused lines (`linea_no_utilizada`) at 31 December 2015. Revolving status is the extract's revolving field; "not identifiable" means the account code does not reveal it. The class column is the product-based classification used in the regressions. "Other lines" does not appear because that product was introduced in June 2018. Subtotals are sums of the rounded entries.

6 Regression Specification

6.1 The regression

$$y_{f,b,c,t} = \alpha_{f,c,t} + \beta_c X_{b,t} + \varepsilon_{f,b,c,t}$$

where $y_{f,b,c,t}$ is loan growth from Equation (2), X_{bt} is the bank shock from Equation (1), and $\alpha_{f,c,t}$ is a firm-by-date fixed effect. $c \in \{\text{line}, \text{other}\}$ Standard errors are clustered by bank.

The estimation sample keeps only firms with at least two banks in a given date-credit type.

6.2 Loan growth

The unit of observation is a relationship: bank b, firm i, credit type $k \in \{\text{line}, \text{other}\}$, month t. The balance B_{bikt} sums over all products and currencies within the relationship. The outcome is

$$y_{bikt} \;=\; \log B_{bikt} - \log B_{bik,t-1}. \tag{2}$$

The outcome measures growth in *continuing* relationships. It does not capture new relationships or relationships that end. Foreign-currency balances are in soles, so growth in the ME part includes exchange-rate revaluation.

Growth is calculated in log changes for a few reasons: standard in the literature; makes the proportionality assumption underlying the KM fixed effect more plausible (since we assume that β is homogeneous across banks and firms, having the log on the LHS makes it more plausible that a shock to a bank has the same proportional effect on loans of different sizes); gives us a simple interpretation of the difference between the coefficients for the two credit types (see Section 6.3).

6.3 What the coefficient estimates

The variable of interest is the bank shock X_{bt}. The coefficient β_c is the change in a relationship's log growth for a one percentage point larger deposit-weighted increase in the reserve rate. It is identified from differences *between banks within a firm-month*. Take a firm that borrows from banks b and b' in month t. The coefficient compares growth at b and b', given that their shocks differ.

Three consequences follow.

It is a relative effect. Anything that moves all banks together is absorbed by the fixed effect. This includes the aggregate effect of the policy change. The coefficient says how much more the more-exposed bank cuts, not how much lending falls overall.

The sign is clear under the hypothesis. A tightening reduces the loanable funds of the exposed bank. A supply-side response means a negative coefficient. A positive or zero coefficient means no differential supply response in this sample.

Credit lines and other credit are estimated separately. The comparison of β^j_{line} with β^j_{other} shows whether the response differs across loan types.

Let B_c be the bank supply of credit of type $c \in \{\text{Term}, \text{Line}\}$

$$\beta_c = \frac{dB_c}{dX}\frac{1}{B_c} \Rightarrow \frac{d\left(\frac{B_L}{B_T}\right)}{dX} = \frac{B_L}{B_T}(\beta_L - \beta_T)$$

That is, the difference between the β's is the percentage effect on the relative bank supply of credit lines to term loans of a unit increase in the shock (one percentage point increase in the reserve requirement rate).

6.4 Why this specification

Firm-by-date fixed effects. The central problem in credit-supply work is that firms also change their demand. A firm hit by a bad shock borrows less from every bank. Comparing banks within the same firm and month removes this. This is the approach of Khwaja and Mian (2008) for controlling for firm credit demand, under the assumption that demand shocks lead to a symmetric change in credit demand with each bank relationship. is it essential? maybe not

Credit lines versus other credit. The project studies the supply of credit lines. Reserve requirements change a bank's liquidity cost. Credit lines are the products that let a firm draw on a commitment. Other credit is a natural comparison group. Running the same regression on both lets the data show whether the response is specific to lines.

6.5 Identifying assumptions and threats

The coefficient has a supply interpretation under three assumptions.

Common demand within firm-month. A firm's demand for credit is the same at all of its banks. This is the Khwaja–Mian assumption. It can fail if firms direct their demand to specific banks, for example by currency.

Exposure unrelated to other shocks. Banks with different deposit currency mixes must not be hit differently by other events that coincide with the rate changes. The leading concern is the exchange rate. It moves the soles value of foreign-currency balances. It also affects dollarized banks and dollarized borrowers differently. Because the outcome is in soles, relationships with more foreign-currency credit mechanically show larger swings when the exchange rate moves. The shock is a function of deposit dollarization. Any other channel that works through dollarization is therefore confounded with it. The table does not control for the currency mix of the loan.

Predetermined exposure. The December 2009 shares must not predict later bank-specific loan growth in other ways. A fixed pre-period share is the usual defense. It cannot rule out that dollarized banks are different in ways that matter for lending.

need a better understanding of shift-share to say what conditions are sufficient vs necessary

7 Clustering

Clustering by bank. The shock varies at the bank-by-month level. Errors are likely to be correlated within a bank across firms and months. So the standard errors are clustered by bank. This gives 8 or 11 clusters, which is too few for the usual asymptotic approximations.

Wild cluster bootstrap. With few clusters, conventional cluster-robust tests over-reject. The table therefore reports p-values from a wild cluster bootstrap (Cameron, Gelbach and Miller, 2008; Roodman et al., 2019). The null is imposed and there are 9,999 draws. The weights are Webb six-point weights, because with fewer than 12 clusters the Rademacher distribution has too few distinct draws. (All four regressions have fewer than 12 clusters, so all use Webb weights.) The stars in the table come from the conventional clustered $t(G-1)$ test. The bootstrap p-value is the number to rely on.

8 Results

Table 3 gives the results.

Table 3: Reserve-requirement shocks and loan growth: credit lines versus other credit

Shock measure	Minimum legal rate		Marginal rate	
Credit type	Credit lines	Other credit	Credit lines	Other credit
	(1)	(2)	(3)	(4)
Bank shock	-1.055	0.082	-0.120	0.048
	(0.770)	(0.138)	(0.113)	(0.056)
Banks (= bank clusters)	8	11	8	11
Firms	130	394	95	312
Firm-by-date cells	4,191	17,760	1,693	7,187
Wild cluster bootstrap p-value	0.560	0.404	0.251	0.565
Mean of loan growth	-0.015	-0.019	-0.009	-0.017
Observations	9,230	52,029	3,686	20,921
R^2	0.482	0.352	0.495	0.368
Within R^2	4.56×10^{-5}	2.23×10^{-6}	4.16×10^{-5}	5.23×10^{-5}
Firm × date fixed effects	✓	✓	✓	✓

Notes: The dependent variable is the log change in a firm's credit balance (*crédito vigente*) at a bank between consecutive month-ends, $\log B_{bit} - \log B_{bi,t-1}$, so it measures growth among continuing relationships with strictly positive balances at both dates; loan entry and exit are not captured. Balances are summed across products and currencies within a bank–firm–credit-line status (foreign-currency balances are reported in soles at the month-end exchange rate, so growth includes exchange-rate revaluation). Columns (1) and (3) use credit lines (credit cards, current accounts, letters of credit, guarantees, multi-lines, other lines); columns (2) and (4) use all other credit. Because the balance is performing drawn credit, letters of credit, guarantees, multi-lines and other lines have a zero balance in every record, so the credit-line samples consist of credit cards and current accounts. The bank shock is $s^{mn}_{b,0}\Delta r^{mn}_t + s^{me}_{b,0}\Delta r^{me}_t$, where Δr is the month-end change, in percentage points, of the statutory minimum legal reserve rate (columns 1–2) or marginal reserve rate (columns 3–4) on domestic (mn) and foreign (me) currency obligations (positive = tighter), and $s_{b,0}$ are the bank's fixed December 2009 deposit shares in each currency. The marginal-rate shock is undefined when the domestic regime is a single rate, which shortens columns (3)–(4). Each regression keeps firm-by-date cells with at least two banks and absorbs firm-by-date fixed effects. Standard errors, in parentheses, are clustered by bank; with few clusters the p-value from a wild cluster bootstrap (null imposed, 9,999 draws) is reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ from the conventional clustered $t(G-1)$ test.

8.1 Number of Valid Observations

Table 4 shows how the regression samples arise from the raw file.

Two steps shrink the sample sharply. The first is the multiple banks restriction. Most relationship-

Table 4: From the raw credit file to the regression samples (number of rows)

	Credit lines	Other credit
Raw rows (firm × bank × currency × segment × product × month)	86,101	180,929
Aggregated rows (bank × firm × credit type × month)	65,185	120,267
less: first observation of a relationship	995	2,554
less: reporting gap (previous month missing)	1,279	5,050
less: zero balance at t or $t-1$	30,861	23,701
Rows with valid growth	32,050	88,962
Minimum legal rate (columns 1 and 2)		
with a defined shock	31,974	79,133
in firm-by-date cells with at least two banks	9,230	52,029
Marginal rate (columns 3 and 4)		
with a defined shock	12,334	31,113
in firm-by-date cells with at least two banks	3,686	20,921

months drop out here. The second is the marginal shock. It exists only when the MN regime is base plus marginal. That is 67 of the 159 months in which growth can be computed: October 2010 to January 2014 and October 2021 to December 2023. (September 2021 is lost because the change in the MN marginal rate has no starting value.) A side effect: the large ME marginal-rate changes of 2015 to 2018 do not enter columns (3) and (4). The MN marginal change is undefined in those months, so the shock is too.

8.2 Why power is low

Four features of the design limit precision.

Few shock months. In the minimum-legal-rate regressions, the shock is non-zero in 16 of 159 months. In the marginal-rate regressions it is non-zero in 7 of 67 months. All seven of those months fall between October 2010 and January 2014. The October 2021 to December 2023 months add observations but no shock variation.

Few banks. Eight or eleven clusters. The shock varies across banks only through differences in deposit shares, and the large banks have similar shares (about 35–50% MN). The cross-bank spread in the shock is therefore narrow.

Few usable cells. Only 9% to 12% of firm-by-date cells have shocks that differ across banks. In the other cells all banks have the same shock (typically zero). These cells add nothing to β.

Thin credit-line sample. The aggregated credit-line panel has 558 firms and 14 lenders. After the restrictions, 8 banks, 130 firms and 4,191 cells remain. Cells have 2.2 banks on average.

9 Limitations and open items

1. **Drawn balances only.** The outcome is performing drawn credit. Undrawn commitments (`linea_no_utilizada`) are in the file but are not used. Growth in a credit line's drawn balance reflects both the limit and the firm's utilization.

2. **Credit lines are cards and current-account advances.** The other four line products have no drawn balance. The credit-line sample is not representative of all credit lines.

3. **Lines inside "loans".** Revolving lines booked under the loan product cannot be identified and sit in other credit. In an exploratory December 2015 cross-section of a larger extract (`muestra_anonima_base2015_v2.parquet`), the loan product with no identifiable revolving status held about 62% of unused line amounts and 31% of drawn balances (`output/tables/product_table.csv`). This blurs the contrast between the two groups.

4. **Intensive margin only.** Entry and exit of relationships are not captured. If tightening mainly ends relationships, the table misses it.

5. **Currency and exchange rate.** The currency basis of balances is an open question in the data dictionary. If balances are in soles, ME growth includes revaluation. The table has no control for the loan's currency mix.

6. **Outliers.** Loan growth is not winsorized. The 1st and 99th percentiles are about -5.0 and $+4.7$ log points for credit lines (see Table 7).

7. **Timing.** The shock is contemporaneous. There are no lags or leads, and no event-study. Reserve-requirement changes may affect lending with a delay.

8. **Marginal-rate shock.** It ignores each bank's base level and deposit growth. It is an exposure proxy, not the bank's actual reserve cost.

9. **Sample.** The regressions use a 1,000-firm extract. Eleven lenders, holding 80.8% of the balance, have a shock. Banco de la Nación is excluded.

10. **Inference.** With 8 and 11 clusters, the wild bootstrap is needed. The bootstrap uses Webb weights and a fixed seed (20261009).

References

Khwaja, Asim Ijaz and Atif Mian, "Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market," *American Economic Review*, September 2008, *98* (4), 1413–42.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.

Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in Stata using `boottest`. *The Stata Journal*, 19(1), 4–60.

A Supporting tables

Table 5: Banks in the regressions: baseline deposit shares and observations

Bank	Dec 2009 deposit share		Credit file		Regression rows	
	MN	ME	Firms	Rows	Lines (col. 1)	Other (col. 2)
Banco de Crédito (BCP)	0.405	0.595	791	94,622	3,585	13,184
BBVA Perú	0.494	0.506	415	51,242	3,105	10,793
Scotiabank Perú	0.401	0.599	387	48,218	924	11,655
Interbank	0.438	0.562	167	15,795	736	5,572
BIF	0.406	0.594	134	13,863	532	3,009
Mibanco	0.793	0.207	132	5,611	0	2,007
Banco Pichincha	0.503	0.497	79	5,572	70	2,149
Citibank	0.366	0.634	36	4,381	275	1,628
Santander Perú	0.353	0.647	43	3,456	0	1,820
Bancom	0.703	0.297	14	812	3	209
Alfin Banco	0.952	0.048	2	9	0	3

Notes: Deposit shares are MN and ME deposits as shares of total deposits at December 2009, from the SBS deposit series. "Credit file" counts firms and rows for the bank in the raw credit file. "Regression rows" are the rows used in columns (1) and (2) of Table 3. Column (3) and (4) samples are subsets of these.

A.1 Estimation samples

Table 7 describes the four estimation samples. The rows on the shock matter most for interpretation.

B Where each piece is produced

Run the scripts from the project root, in the order 09, 10, 11.

C Linking lenders across datasets

The credit file identifies lenders by RCC entity code. The deposit and rate data use institution names. The link runs through a common key, `bank_standardized`:

$$\text{credit row} \rightarrow \texttt{entidad} \rightarrow \texttt{bank_standardized} \rightarrow \text{bank shock.}$$

The first step uses the RCC entity list. The second uses the hand-curated bank map. Both steps are many-to-one, and a script check confirms that no credit row is duplicated or lost in the merge.

Not every lender has a shock. Table 9 sorts the 83 lender codes in the credit file by the reason they are included or excluded. Eleven lenders have a shock. They hold 80.8% of the summed performing balance in the file. The largest excluded lender is Banco de la Nación (about 8% of the balance), which has no match in the project crosswalks. Most of the rest is portfolio-sale and trust vehicles.

Table 6: Months with a non-zero reserve-rate change in the regression window

	Minimum legal rate		Marginal rate	
Month-end	MN	ME	MN	ME
Oct 2010	+0.5	+0.5	+10	+5
May 2012	0	0	+5	0
Aug 2013	0	0	−5	−5
Sep 2013	0	0	−5	0
Oct 2013	0	0	−4	0
Dec 2013	0	0	−1	0
Jan 2014	0	0	−1	0
Jan 2015	0	0	–	+10
Feb 2015	−0.5	0	–	0
Mar 2015	−0.5	0	–	+10
Apr 2015	−0.5	0	–	0
May 2015	−0.5	0	–	0
Jun 2015	−0.5	0	–	0
Jan 2017	−0.5	0	–	−22
Mar 2017	0	0	–	−2
Apr 2017	−1	0	–	0
May 2017	0	0	–	−2
Jul 2017	0	0	–	−2
Sep 2017	0	0	–	−1
Oct 2017	0	0	–	−1
Jan 2018	0	0	–	−1
Feb 2018	0	0	–	−1
Mar 2018	0	0	–	−1
Apr 2018	0	0	–	−1
Jul 2018	0	0	–	−1
Apr 2020	−1	0	–	0
Nov 2021	+0.5	0	0	0
Dec 2021	+0.25	0	0	0
Jan 2022	+0.25	0	0	0
Feb 2022	+0.25	0	0	0
Mar 2022	+0.25	0	0	0
Apr 2022	+0.25	0	0	0
May 2022	+0.25	0	0	0

Notes: Month-end change in the rate, in percentage points ($\Delta r_t = r_t - r_{t-1}$; positive = tighter). "–" means the change is undefined because the MN marginal rate does not exist under the single-rate regime (February 2014 to August 2021) or in the month the regime returns (September 2021). Months not listed have no change in any of the four rates. Window: October 2010 to December 2023.

Table 7: Estimation samples: outcome and shock variation

Shock measure	Minimum legal rate		Marginal rate	
Credit type	Lines	Other	Lines	Other
	(1)	(2)	(3)	(4)
Observations	9,230	52,029	3,686	20,921
Months in sample	159	159	67	67
Mean banks per firm-by-date cell	2.20	2.93	2.18	2.91
Loan growth				
Loan growth, mean	−0.015	−0.019	−0.009	−0.017
Loan growth, std. dev.	1.370	0.757	1.375	0.794
Loan growth, 1st percentile	−5.010	−1.624	−4.907	−1.486
Loan growth, median	−0.030	−0.018	−0.015	−0.024
Loan growth, 99th percentile	4.700	1.869	4.562	1.887
Bank shock				
Months with a non-zero shock	16	16	7	7
Share of observations with non-zero shock	9.4%	10.1%	14.7%	11.6%
Share of firm-by-date cells with shock variation across banks	8.9%	9.6%	12.2%	9.9%
Std. dev. of shock, pooled	0.078	0.082	1.245	1.148
Std. dev. of shock, within firm-by-date cell	0.006	0.011	0.052	0.095

Notes: Statistics on the estimation samples of Table 3. "Within cell" is the standard deviation of the shock after subtracting the firm-by-date cell mean; this is the variation that identifies the coefficient. Loan growth is not winsorized.

Table 8: Scripts and outputs

Script	Output
`09_encaje_series.R`	`data/processed/encaje_rates_monthly.csv`: monthly reserve-rate panel rebuilt from BCRP circulars.
`03_validate_direct.R`	`crosswalks/xwalk_reporte.csv`: map from deposit-series institution names to `bank_standardized`. Needed by script 10.
`10_reserve_shocks.R`	`bank_deposit_shares.csv` (shares, all institutions and months); `bank_encaje_shocks.csv` (shock by bank and month, with the December 2009 baseline).
`11_loan_reserve_shock_regression.R`	`credit_line_growth_panel.csv.gz` (relationship-month panel with growth and shocks); `output/tables/tab_encaje_loan_growth.tex`; diagnostics in `output/validation/encaje_shock_*.csv`.
`12_figures_regression.R`	Figures of the reserve rates and of deposit shares over time.

Table 9: Lender coverage in the credit file

Status	Lender codes	Rows	Share of balance
Included: valid bank shock	11	243,581	80.8%
Excluded: not matched to any bank in the project crosswalks (includes Banco de la Nación)	10	2,786	11.3%
Excluded: portfolio-sale or trust vehicle (RCC status "Venta de Cartera")	20	8,704	7.5%
Excluded: bank without valid December 2009 deposit shares	9	2,645	0.3%
Excluded: entity code absent from the RCC entity list	18	1,921	0.0%
Excluded: cooperative or municipal / rural savings bank (outside the Reporte)	15	7,393	0.0%

Notes: Share of balance is the lender's share of the sum of `credito_vigente` over all rows of the raw credit file. The 11 included lenders are the banks listed in Table 5. Eight further institutions have valid December 2009 shares but no matching lender in the credit file.